On the Faithfulness of Language Model-based Data-to-Text Generation

[PhD Thesis Presentation]

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Outline

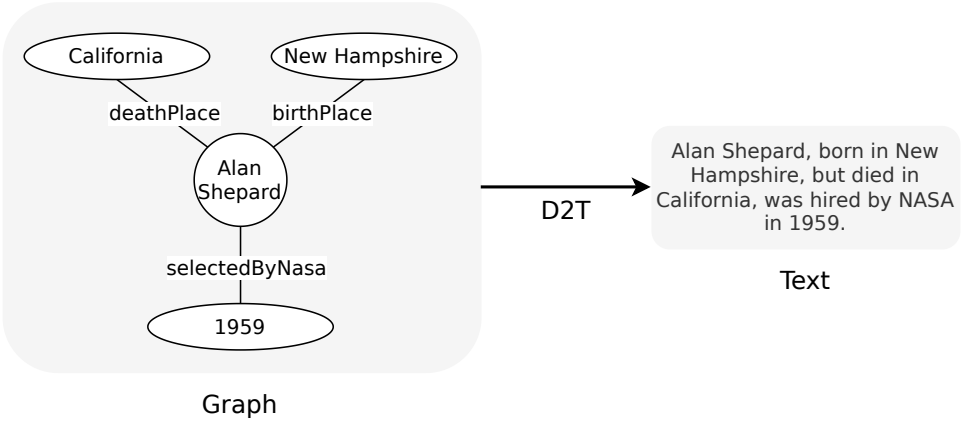
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Data-to-Text Generation (D2T)

Data-to-text generation (D2T) aims to produce textual descriptions of semi-structured data such as tables, graphs (or sets of relational triples), and slot–value pairs (meaning representations or MRs).



Alan Shepard.



An example of data/graph-to-text generation.

Applications of D2T. Dialogue systems, weather forecasting, automated journalism, medical report writing, . . .

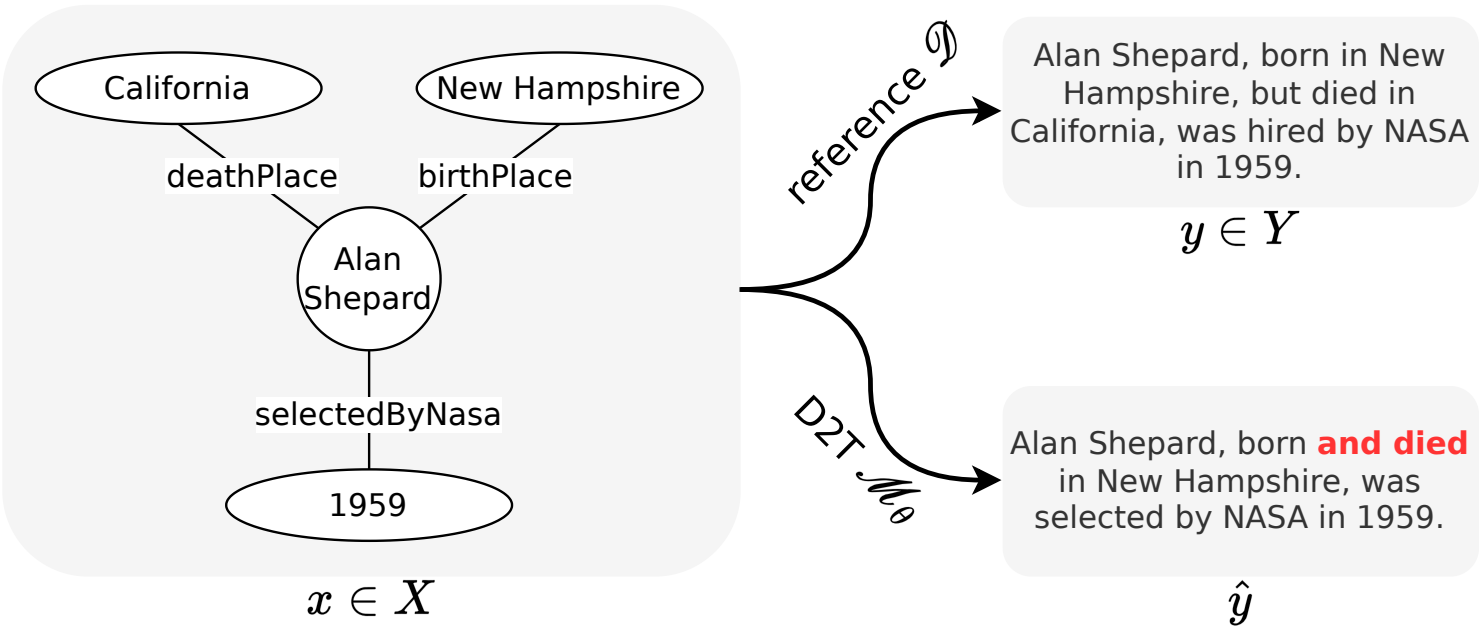
An Engineering Perspective on D2T

D2T can be expressed as a function

$$\mathcal{F} : X \rightarrow Y$$

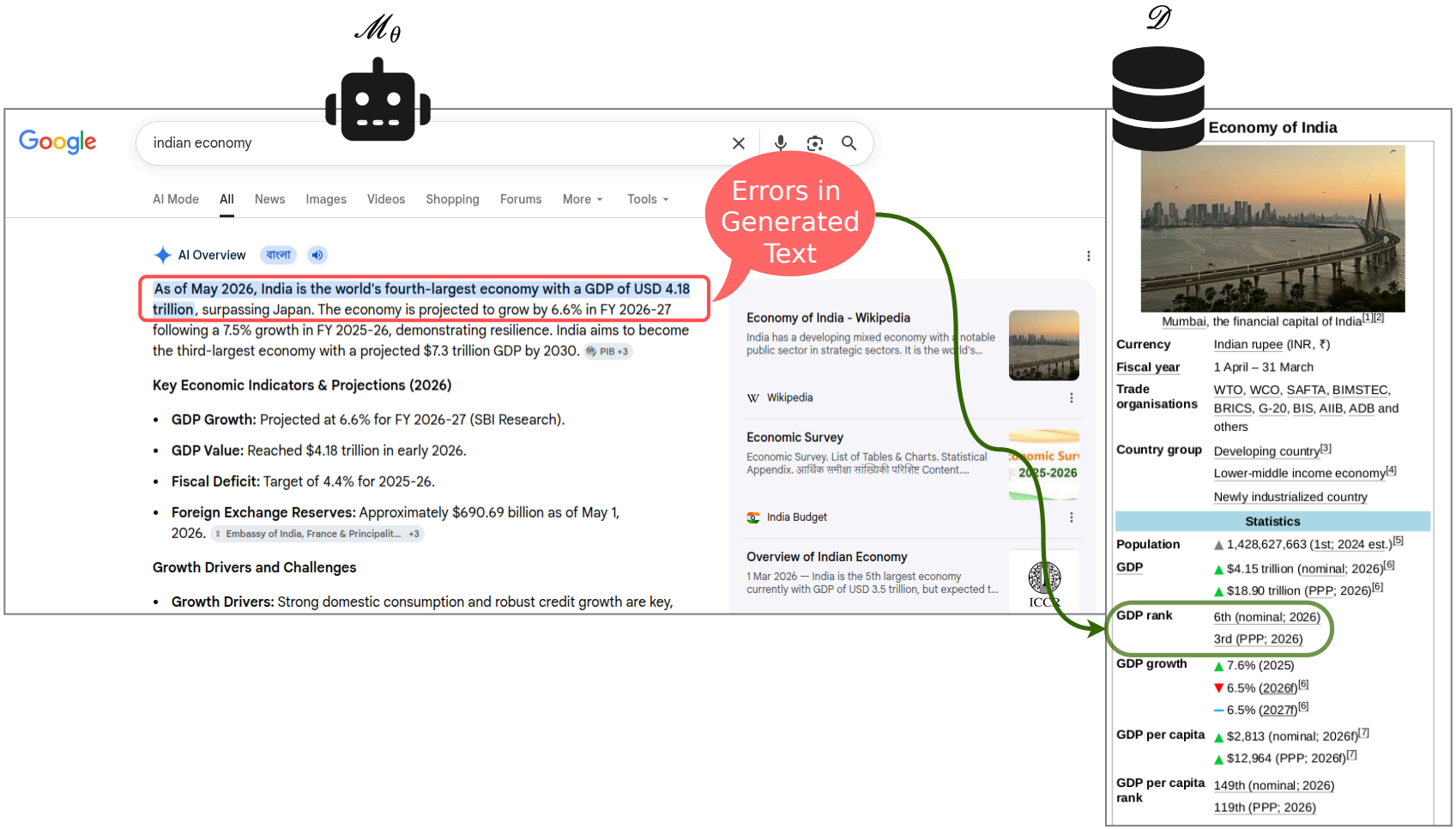
- $x \in X$, where x represents the *context/source*, and X denotes the space of structured/semi-structured data.
- $y \in Y$, where y represents the *hypothesis/generated text*, and Y denotes the space of natural language texts.
- This mapping is parameterized by a model $\mathcal{M}_\theta \approx \mathcal{F}$. Modern approaches typically use a language model for $\mathcal{M}_\theta$.
- **Training.** The parameters θ are optimized to maximize the likelihood over a training dataset $\mathcal{D} = \left\{(x^{(i)}, y^{(i)})\right\}_{i=1}^{n}$.
- **Inference.** Through decoding strategies, the generated output is obtained as $\widehat{y} \leftarrow \mathcal{F}(x)$.

A Closer Look

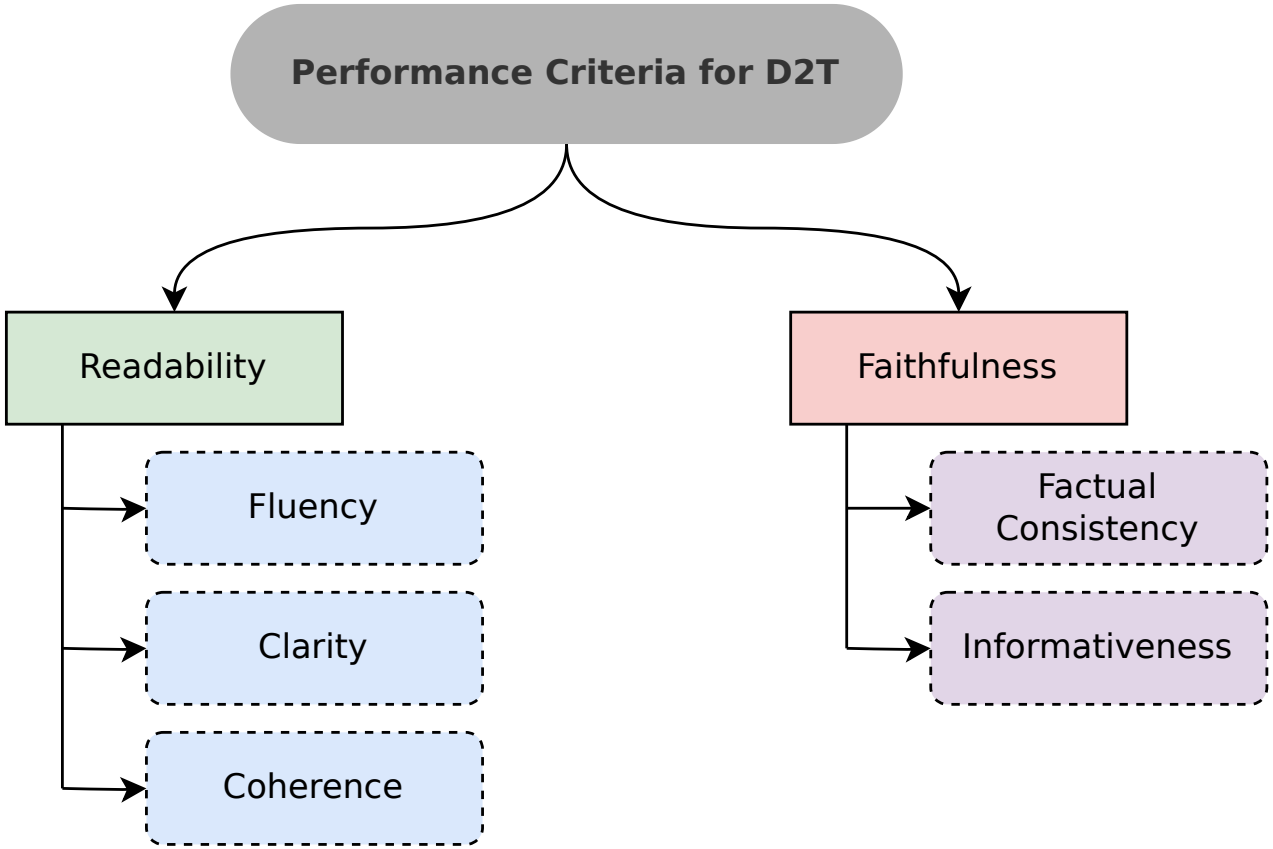


A closer look at a D2T example, highlighting the generation error in red.

Sometimes, Even Google™ Gets It Wrong!



Performance Measurement of D2T: Readability and Faithfulness



Performance Measurement of D2T: Readability

Readability measures how closely the generated text ($\hat{y}$) aligns with human language norms.

- **Fluency:** Grammatical correctness and natural surface realization of the text.
- **Coherence:** Logical and discourse-level organisation that yields a smooth progression of ideas.
- **Clarity:** Precision and lack of ambiguity in expression so that the reader can unambiguously interpret the intended meaning.

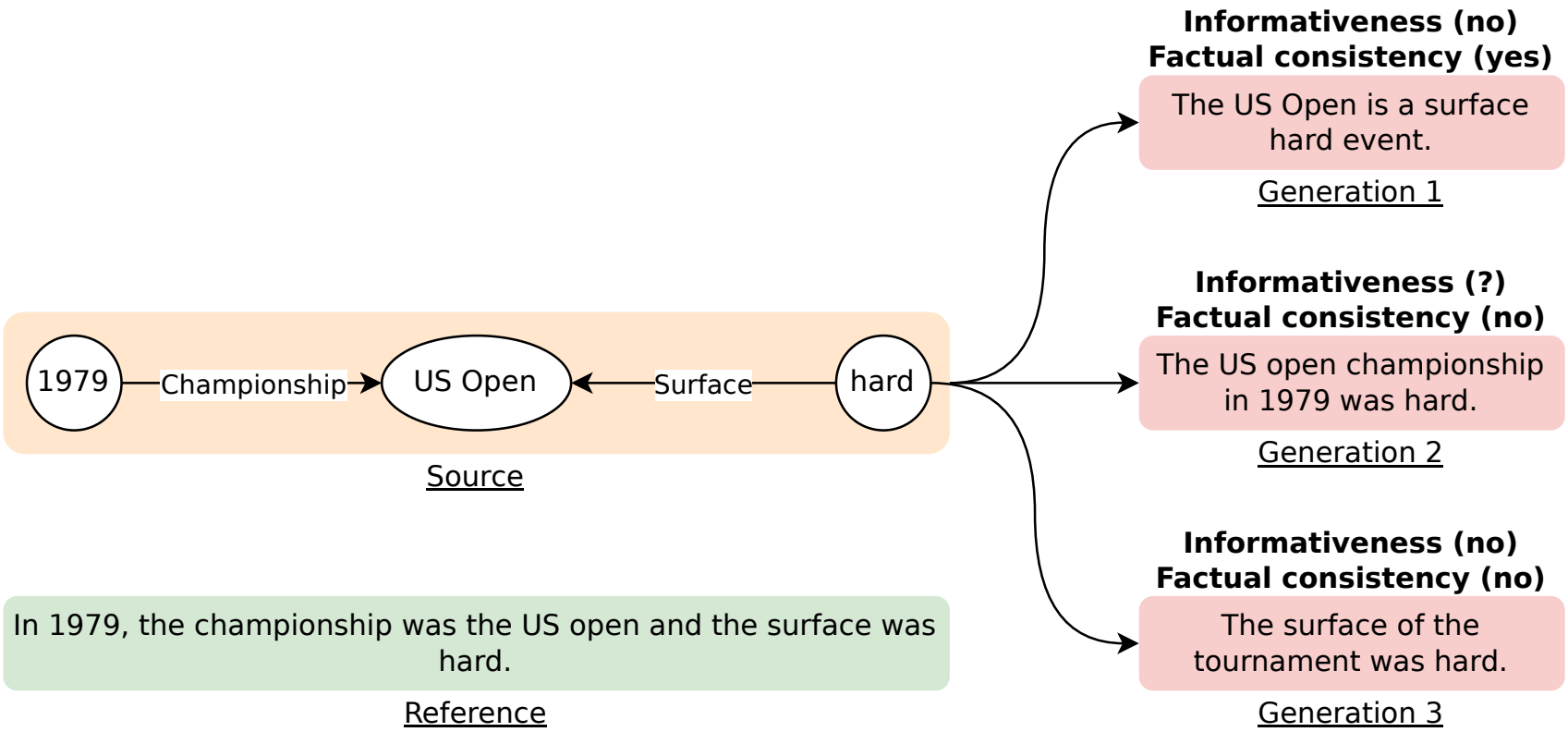
Language model-based D2T systems, particularly those using large language models (LLMs), exhibit exceptional performance in terms of readability.

Performance Measurement of D2T: Faithfulness

Faithfulness is a key measure in D2T, assessing the extent to which the generated text ($\hat{y}$) accurately reflects the factual content, emphasis, and salience of the source input (x).

- **Informativeness:** The generated text should include all salient and relevant information from the given input, reflecting its key facts and emphasis.
- **Factual Consistency:** Every statement in the generated text must be factually correct with respect to the input and should not introduce unsupported or contradictory information.

Examples of Unfaithful Generation in D2T



Research Challenges

⚠ Language model-based D2T systems, particularly those powered by large language models (LLMs), often fall short in terms of **faithfulness**.

- Lack of informativeness leads to amnesic behavior.
- Lack of factual consistency results in hallucination.

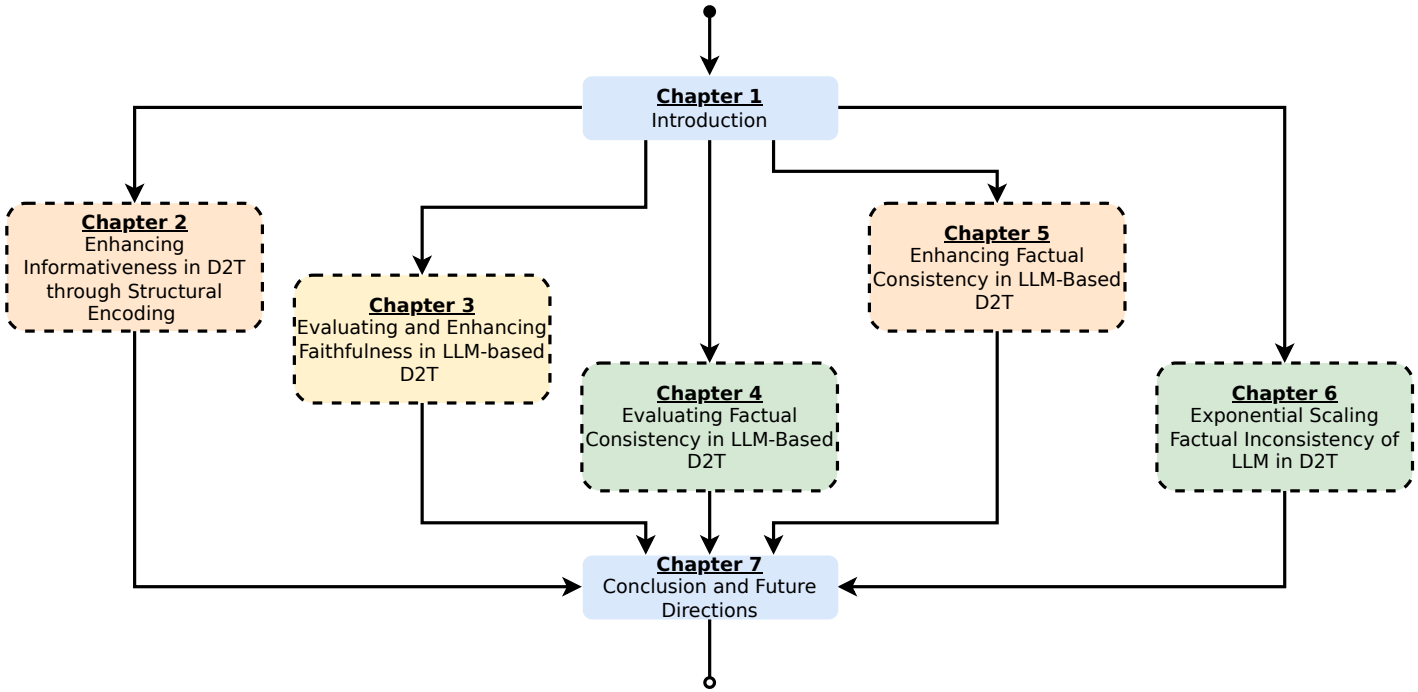
Why Faithfulness Matters in D2T

Faithfulness is vital in safety-critical applications such as healthcare, finance, and climate reporting, where unfaithfulness can have serious real-world consequences.

Thesis Objectives

The primary objectives of this thesis are twofold:

- **Evaluation:** To systematically assess the *faithfulness* of D2T, focusing on its two core dimensions — *informativeness* and *factual consistency*.
- **Enhancement:** To develop and integrate methods that improve model faithfulness across these dimensions.



Thesis Work 1

Enhancing Informativeness in D2T through Structural Encoding

Mahapatra, J. and Garain, U., 2021. Exploring Structural Encoding for D2T. *International Conference on Natural Language Generation (INLG)*, Association for Computational Linguistics.

Thesis Work 1: Background and Objective

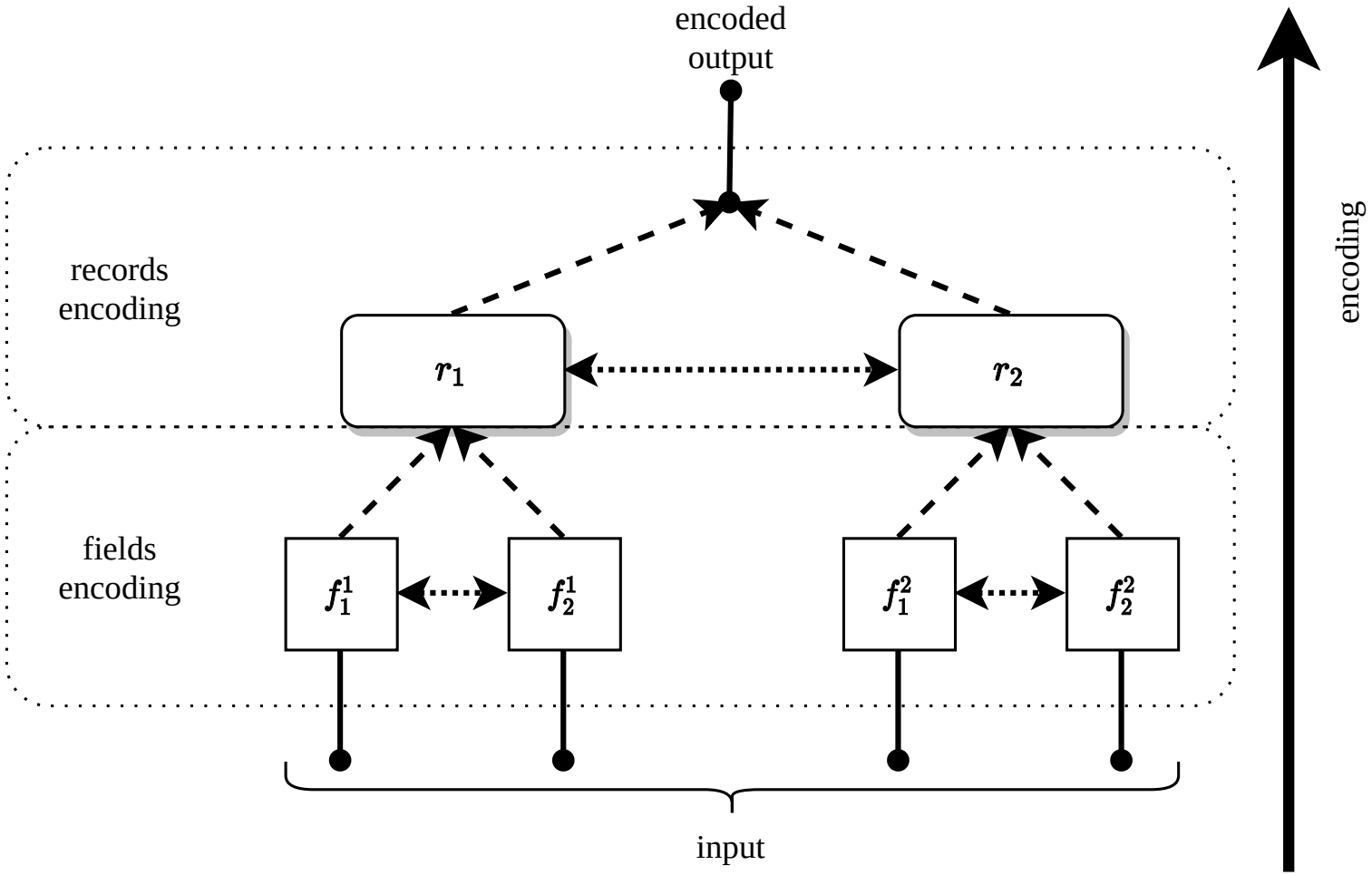
Background (Pre-LLM Era)

- Encoder–decoder models are widely used in D2T for their efficiency and fluency, yet producing informative input encodings remains challenging.
- D2T inputs typically follow a **record-field structure**.
- Existing methods emphasize attention mechanisms over capturing underlying **structural dependencies**.

Objective

To develop structure-aware encoding methods that effectively capture record-field relationships of source data for improved content selection (informativeness) and overall faithfulness.

Structural Encoding



Horizontal connections

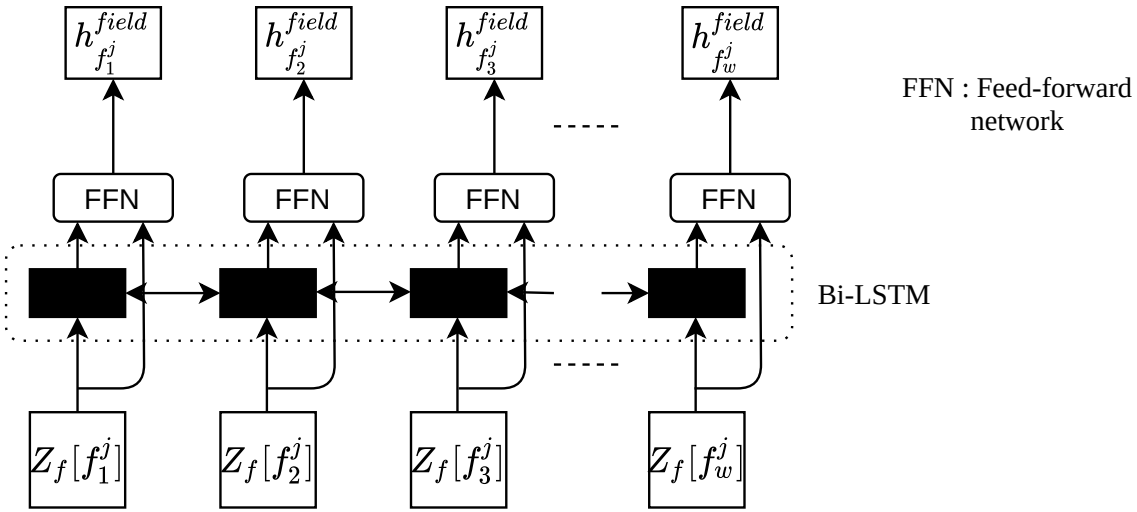


Figure: Horizontal connections in field encoding.

Hierarchical connections

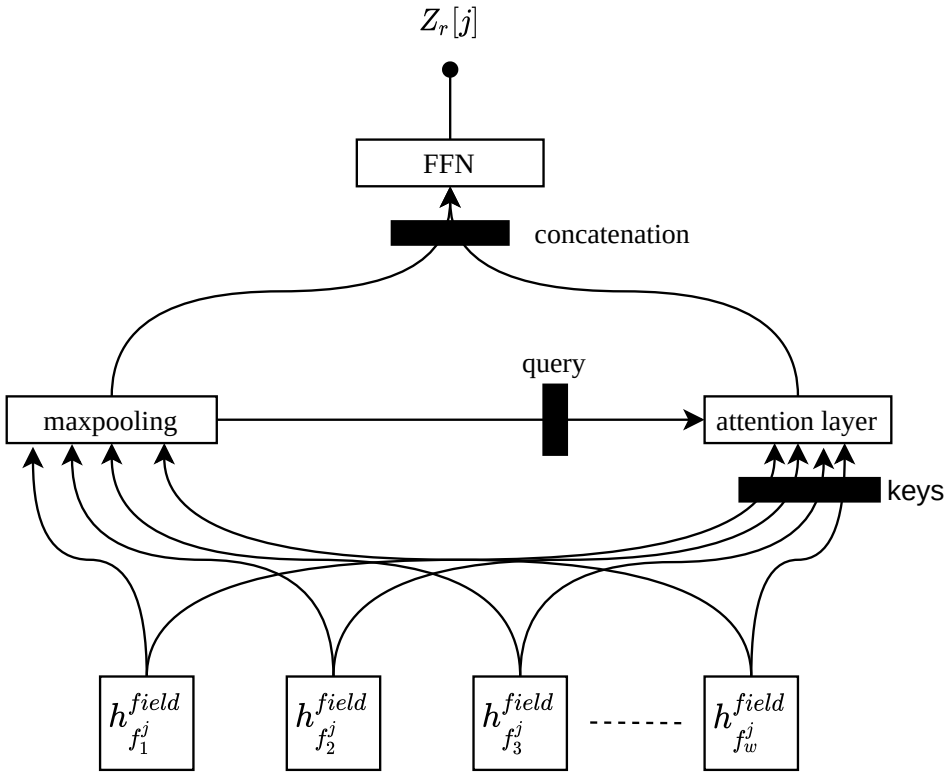
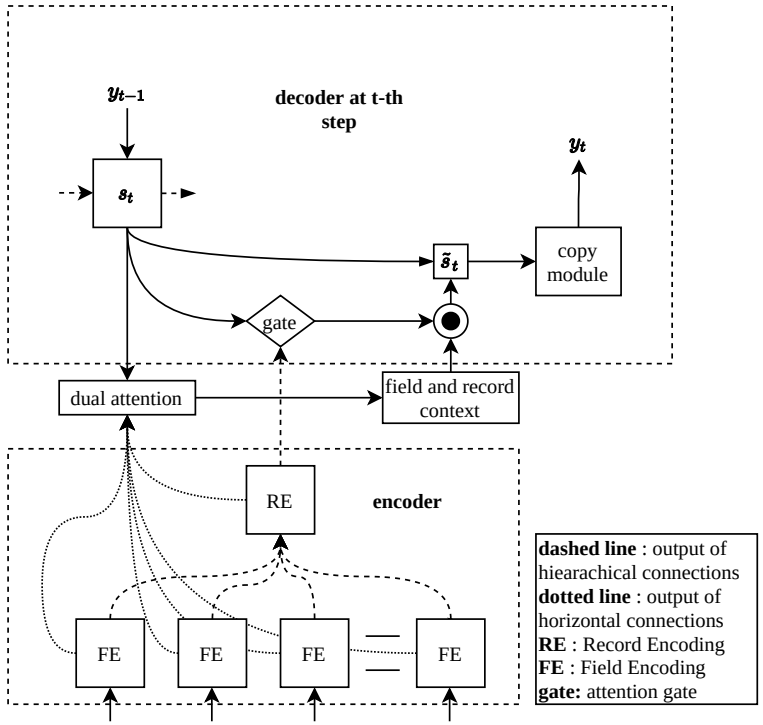


Figure: Hierarchical connections in field encoding.

Attention gates (Content Selection)

$$\text{Record Attention gate: } g_t^r = \sigma(\mathbf{W_{rg}} s_t + \mathbf{U_{rg}} Z_d[x] + \mathbf{b_{rg}})$$

$$\text{Field Attention gate: } g_t^f = \sigma(\mathbf{W_{fg}} s_t + \mathbf{U_{fg}} Z_d[x] + \mathbf{b_{fg}})$$



A Snapshot of Findings

Table: Quantitative results from the WikiBio Dataset (PARENT denotes the informativeness).

Model	BLEU	ROUGE-L	METEOR	PARENT
Vanilla Seq2Seq	0.338	0.418	0.271	0.463
Proposed method (without attentions)	0.348	0.445	0.262	0.455
Transformer (from scratch)	0.381	0.486	0.340	0.407
Liu et al. (2018)	0.447	0.528	0.363	0.538
Nie et al. (2019)	0.450	0.522	0.371	0.527
Proposed method	**0.465**	**0.566**	**0.397**	**0.540**

Table: Ablation of attention gates.

	WikiBio		E2E	
	BLEU	PARENT	BLEU	PARENT
with attention gates	**0.465**	**0.540**	0.674	**0.716**
without attention gates	0.458	0.513	**0.680**	0.694

Thesis Work 2




Evaluating and Enhancing Faithfulness in LLM-based D2T

Mahapatra, J. and Garain, U., 2026. A Comprehensive Performance Evaluation of LLMs for D2T and Divergence-Weighted Training. *ACM Transactions on Intelligent Systems and Technology (ACM TIST)*.

Thesis Work 2: Overview

Research Gap

Despite the success of LLMs in other text generation tasks, D2T still lacked a systematic performance evaluation spanning different LLM families, datasets, and model sizes.

Contributions

- Evaluate multiple **LLMs of varying sizes** for D2T across LLM families (BART, T5, OPT, BLOOM, and Llama 2), datasets, and evaluation metrics (BLEU, METEOR, BERTScore, MoverScore, and BARTScore).
- Analyze LLMs' robustness under source–reference divergence.
- Propose a divergence-aware training strategy to improve faithfulness.

A Finding on Faithfulness with BARTScore

Table: Faithfulness evaluation of LLMs using BARTScore.

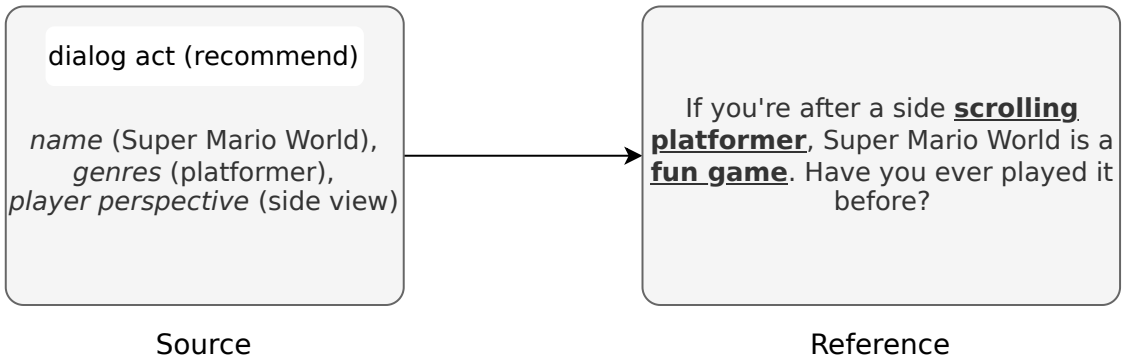
family	model	size (number of parameters in billion)	E2E	ViGGo	WikiTableText	DART	WebNLG
BART	BART-base	0.1	0.086	**0.065**	**0.068**	0.066	**0.119**
	BART-large	0.4	**0.087**	0.064	**0.071**	**0.069**	0.110
T5	T5-base	0.2	0.087	**0.074**	0.070	**0.072**	**0.104**
	T5-large	0.7	**0.093**	0.063	**0.077**	0.070	0.101
OPT	OPT-2.7B	2.7	**0.057**	0.045	**0.068**	**0.063**	**0.089**
	OPT-6.7B	6.7	0.053	**0.046**	0.066	0.062	0.087
	OPT-13B	13.0	0.050	0.043	0.065	0.063	0.087
BLOOM	BLOOM-1.1B	1.1	**0.089**	**0.042**	**0.057**	**0.064**	**0.090**
	BLOOM-3B	3.0	0.083	0.041	0.054	0.064	0.089
	BLOOM-7B	7.0	0.086	0.040	0.056	0.063	0.088
Llama 2	Llama2-7B	7.0	**0.094**	**0.05**	**0.062**	0.073	**0.106**
	Llama2-13B	13.0	0.089	0.048	0.061	**0.074**	0.105

- Increasing **model size** does not consistently improve the **faithfulness** of LLMs.
- No single LLM family dominates across all datasets.

Source–reference divergence

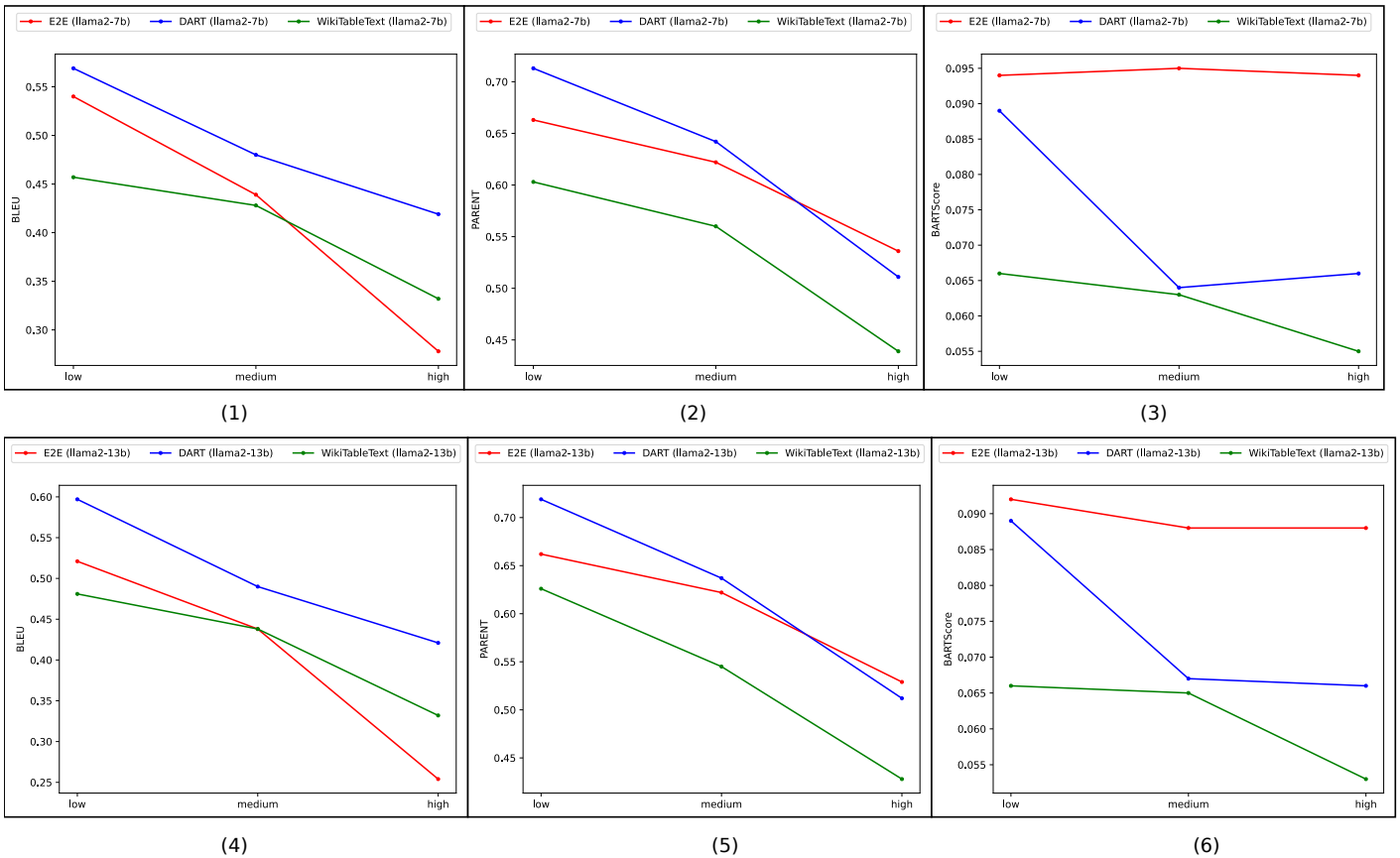
- **Source–reference divergence** denotes the information gap between the input (source) data and its reference text, arising from dataset construction, human bias and task variability.

$$\text{div}(s, r) = 1 - \frac{|\text{unigram}(s) \cap \text{unigram}(r)|}{|\text{unigram}(s) \cup \text{unigram}(r)|}$$



An example of source-reference divergence.

Performance Degradation under Source–reference divergence



Analysis of performance of Llama2 in context of source-reference divergence.

Source-Reference Divergence-Weighted Training

$$\theta^* = \underset{\theta}{\text{argmax}} \sum_{(s,r)\in\mathcal{D}} w(s,r) \cdot \mathcal{L}\big(\mathcal{M}_\theta(s), r\big)$$

where the source is denoted by s, the reference by r, and the associated weight is defined as $w(s,r) = 1 - \text{div}(s,r)$, with $\text{div}(s,r) \in [0,1]$.

- High divergence $\Rightarrow$ lower weight
- Low divergence $\Rightarrow$ higher contribution to training

Outcomes of Divergence-Weighted Training

☻ Performance comparison between conventional and divergence-weighted training.

metric	family	model	size (number of parameters in billion)	conventional training			Weighted training		
				E2E	WikiTableText	DART	E2E	WikiTableText	DART
BLEU	T5	T5-base	0.2	0.380	0.365	0.404	**0.387**	**0.367**	**0.446**
		T5-large	0.7	0.398	**0.390**	0.442	**0.418**	0.388	**0.462**
	BLOOM	BLOOM-3B	3.0	0.380	0.396	0.446	**0.388**	**0.427**	0.442
		BLOOM-7B	7.0	0.379	0.423	0.444	**0.411**	**0.438**	**0.469**
	Llama 2	Llama2-7B	7.0	0.419	0.436	**0.494**	**0.451**	**0.449**	0.439
		Llama2-13B	13.0	0.408	0.451	0.510	**0.448**	**0.516**	**0.528**
PARENT	T5	T5-base	0.2	0.590	**0.472**	0.577	**0.602**	0.470	**0.582**
		T5-large	0.7	0.604	0.512	0.606	**0.620**	**0.545**	0.573
	BLOOM	BLOOM-3B	3.0	0.569	**0.476**	0.556	**0.572**	0.463	**0.582**
		BLOOM-7B	7.0	0.565	0.504	0.557	**0.573**	**0.556**	**0.591**
	Llama 2	Llama2-7B	7.0	0.602	0.531	**0.621**	**0.616**	**0.557**	0.608
		Llama2-13B	13.0	0.599	0.531	0.622	**0.612**	**0.558**	**0.632**
BARTSCORE	T5	T5-base	0.2	0.087	0.070	0.072	**0.088**	**0.075**	**0.081**
		T5-large	0.7	0.093	**0.077**	0.070	**0.104**	0.072	**0.086**
	BLOOM	BLOOM-3B	3.0	0.083	0.054	0.064	**0.085**	**0.061**	**0.076**
		BLOOM-7B	7.0	0.086	0.056	0.063	**0.097**	**0.094**	0.063
	Llama 2	Llama2-7B	7.0	0.094	0.062	0.073	**0.096**	**0.075**	**0.078**
		Llama2-13B	13.0	**0.089**	0.061	0.074	0.088	**0.086**	**0.082**

Thesis Work 3

Evaluating Factual Consistency in LLM-Based D2T

Mahapatra, J. and Garain, U., 2025. A Comprehensive Study on Factual Consistency in LLMs for Data-to-Text Generation. *Sādhanā*, Springer.

Thesis Work 3: Overview

Research Gap

- Prior studies on faithfulness have mainly emphasized **informativeness**, overlooking **factual consistency**.
- The impact of **source–reference divergence** on factual consistency remains under-explored.

Contributions

- Conduct a large-scale and systematic evaluation of factual consistency in LLM-based Data-to-Text (D2T) generation across multiple D2T datasets.
- Assess how model size of LLM influences factual consistency.
- Investigate the impact of source–reference divergence on factual consistency.

Experimental Setup

Datasets

MR-to-Text (E2E, ViGGo), **Table-to-Text** (WikiTableText), and **Graph-to-Text** (DART, WebNLG).

LLM Families

Evaluates twelve open-source **LLMs** from five major families — **BART**, **T5**, **OPT**, **BLOOM**, and **Llama 2** — spanning multiple model sizes.

Evaluation Metrics

Employ a combination of automatic metrics for factual consistency (**SummaC-Conv**, **NEOverlap**, **AlignScore**, **QAFactEval**) and human evaluations.

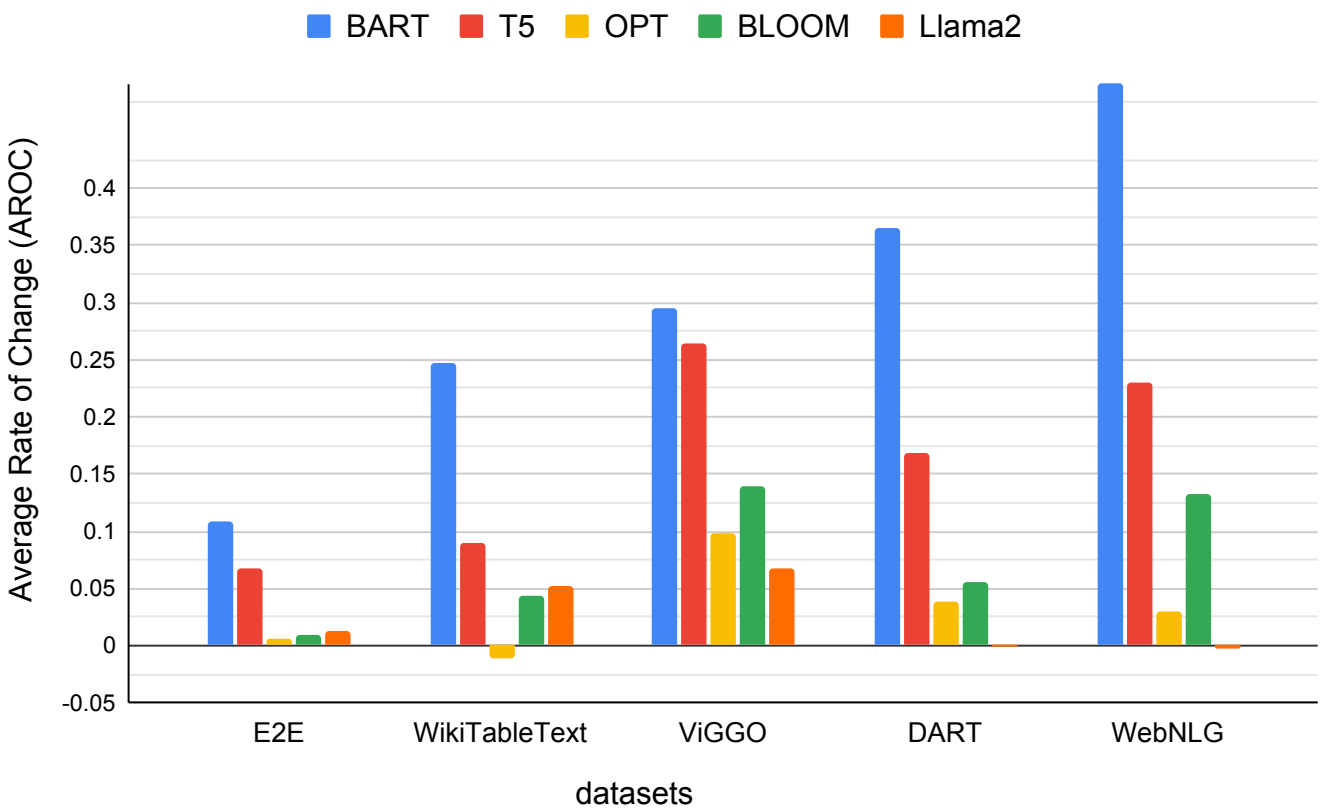
A Snapshot of Finding on Factual consistency

Table: Factual consistency evaluated through QAFACTEVAL.

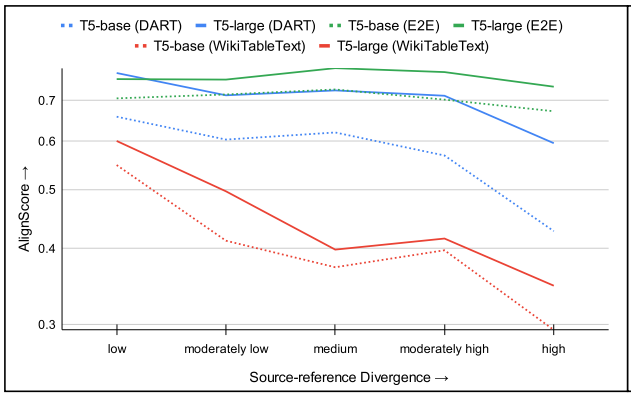
family	model	model size (billion)	E2E	WikiTableText	ViGGo	DART	WebNLG
BART	BART-base	0.14	3.4109	1.8008	1.3894	2.2541	2.1009
	BART-large	0.41	**3.7393**	**2.3559**	**1.9500**	**3.1206**	**3.3009**
T5	T5-base	0.22	3.6086	2.1887	1.6307	2.7631	2.5079
	T5-large	0.75	**3.7393**	**2.6267**	**2.0971**	**3.4325**	**3.3865**
OPT	OPT-2.7B	2.70	3.2732	2.5878	2.4776	3.1245	3.5771
	OPT-6.7B	6.70	**3.2930**	**2.6805**	2.7332	3.1335	3.6794
	OPT-13B	13.00	3.2768	2.6085	**2.8302**	**3.2894**	**3.7640**
BLOOM	BLOOM-1.1B	1.10	3.5579	2.2128	2.0289	3.0540	2.9072
	BLOOM-3B	3.00	3.6179	2.4706	2.5181	3.2751	3.4617
	BLOOM-7B	7.00	**3.6245**	**2.5299**	**2.5265**	**3.3069**	**3.5181**
Llama 2	Llama2-7B	6.00	4.1619	3.0296	2.9628	4.0208	**4.1166**
	Llama2-13B	13.00	**4.1874**	**3.0933**	**3.0950**	**4.0411**	4.0892

Quantification: Variation of Factual Consistency with Model Size

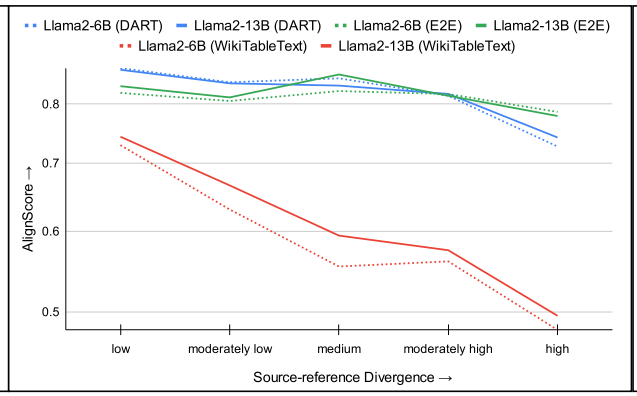
$$\text{Average Rate of Change}\Big|_{\log} = \frac{\log(f(\mathcal{M}_2)) - \log(f(\mathcal{M}_1))}{\log(m_2) - \log(m_1)}$$



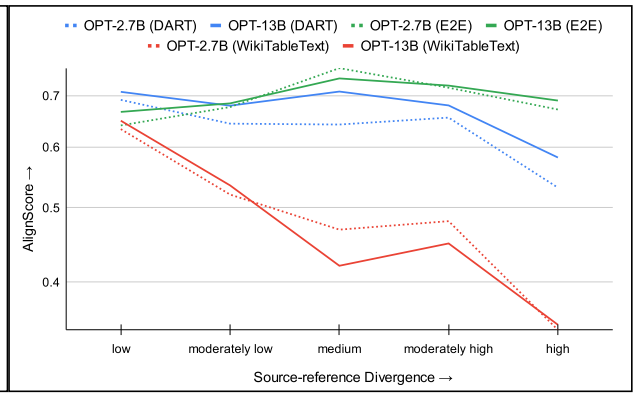
Impact of Source-reference Divergence on Factual Consistency (AlignScore)



(a)



(b)



(C)

Thesis Work 4

Enhancing Factual Consistency in LLM-Based D2T

Mahapatra, J. and Garain, U., 2025. Divergence-Aware Selective Data-to-Text Generation with LLMs for Factual Consistency. *International Conference on Pattern Recognition and Machine Intelligence (PReMI)*.

Thesis Work 4: Overview

From the previous work, a key cause of factual inconsistency was identified as **source–reference divergence**.

Contributions

- Conduct a systematic **analysis of source–reference divergence** across multiple D2T datasets.
- Propose a **divergence-aware selective generation framework** that enables LLMs to abstain from generation when input–reference divergence exceeds a threshold.
- Demonstrate through experiments across multiple datasets and evaluation metrics.

Quantification of Source–Reference Divergence in D2T

- **Unigram-Level Divergence:** Measures lexical disparity between source (x) and reference (y).

$$\text{Unigram}_{\text{div}}(x, y) = 1 - \frac{|\text{unigram}(x) \cap \text{unigram}(y)|}{|\text{unigram}(x) \cup \text{unigram}(y)|}$$

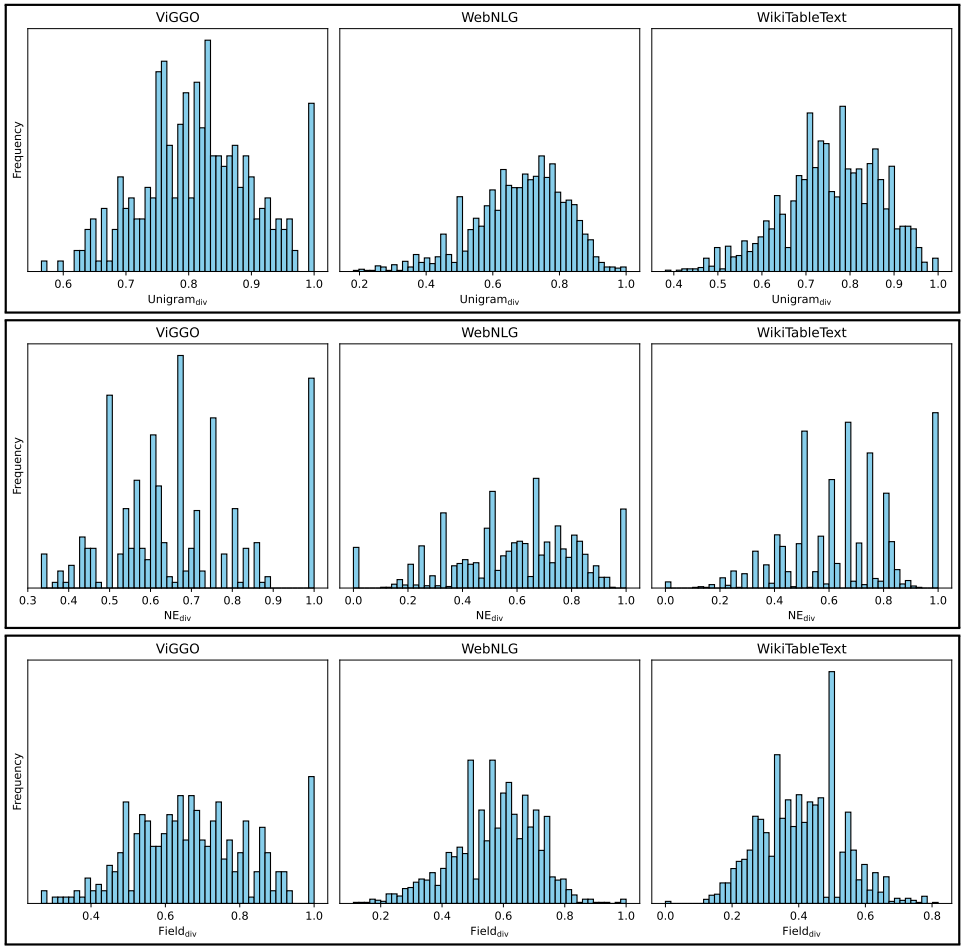
- **Named Entity-Based Divergence:** Captures factual mismatches by comparing named entities.

$$\text{NE}_{\text{div}}(x, y) = 1 - \frac{|\text{NE}(x) \cap \text{NE}(y)|}{|\text{NE}(y)|}$$

- **Field-Aware Divergence:** For structured inputs; evaluates divergence across fields with weights w_f.

$$\text{Field}_{\text{div}}(x, y) = 1 - \frac{1}{|\mathcal{F}_x|} \sum_{f \in \mathcal{F}_x} w_f \cdot \frac{|\text{n-gram}(f) \cap \text{n-gram}(y)|}{|\text{n-gram}(y)|}$$

Existence of Source-Reference Divergence in D2T



Divergence-aware Selective Generation

$$\text{Selective generation} = \begin{cases} \widehat{y} & \text{if } \widehat{y} \sim \mathcal{G}_{\theta*}(x) \text{ and } f_{\text{div}}(x, \widehat{y}) \leq \Phi \\ \text{abstain} & \text{otherwise} \end{cases}$$

🚀 Divergence-aware selective generation on the WikiTableText dataset.

f_{div}	coverage	AlignScore	QAFactEval	Summac-conv	UniEval-fact
NE_{div}	1.0	0.309	0.296	0.183	0.159
	0.9	0.351	0.527	0.410	0.201
	0.8	0.377	0.360	0.243	0.465
	0.7	0.617	0.603	0.482	0.462
	0.6	0.617	0.604	0.480	0.465
$Unigram_{div}$	1.0	0.309	0.296	0.183	0.159
	0.9	0.346	0.525	0.218	0.195
	0.8	0.366	0.597	0.478	0.212
	0.7	0.609	0.600	0.477	0.455
	0.6	0.611	0.605	0.477	0.457
$Field_{div}$	1.0	0.309	0.526	0.183	0.389
	0.9	0.534	0.337	0.220	0.193
	0.8	0.366	0.357	0.478	0.213
	0.7	0.607	0.597	0.483	0.451
	0.6	0.610	0.594	0.483	0.452

Thesis Work 5



Exponential Scaling Factual Inconsistency of LLM in D2T

Mahapatra, J., Roy, S. and Garain, U., Exponential Scaling of Factual Inconsistency in Data-to-Text Generation with Fine-Tuned LLMs. *Transactions on Machine Learning Research (TMLR).*

Thesis Work 5: Overview

Research Gap

Prior scaling studies on text generation through LLMs primarily focus on *generalization error* and *test perplexity* under *power law scaling*, leaving the scaling behavior of **factual inconsistency** largely unexplored.

Contributions

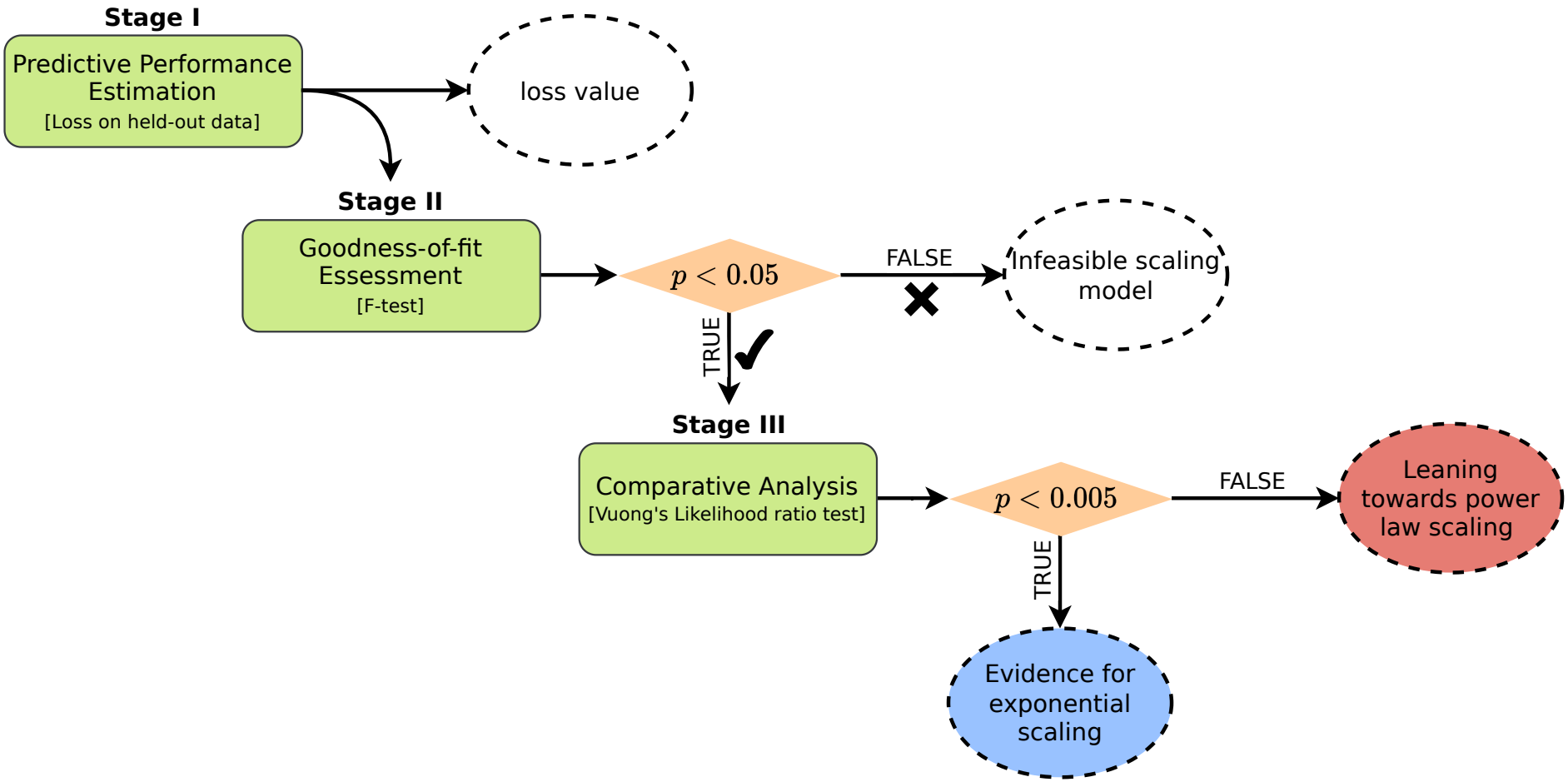
- Provides a comprehensive analysis of how factual inconsistency scales with model size, computational capacity (FLOPs), and fine-tuning data in LLM-based D2T.
- Proposes a *VaCScal* (three-stage statistical validation framework)—to rigorously evaluate scaling laws.
- Demonstrates through extensive experiments across six LLM families and five D2T datasets that factual inconsistency follows an exponential scaling, challenging the prevailing power law scaling.

Desiderata: The Two Scaling Models

$$\text{Power Law Scaling Model} : f(x) = \begin{cases} Ax^{\alpha} + B & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Exponential Scaling Model} : f(x) = \begin{cases} Ce^{\beta x} + D & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

VaCScal (Validation and Comparison of Scaling law)



A Summary (Snapshot) of Results

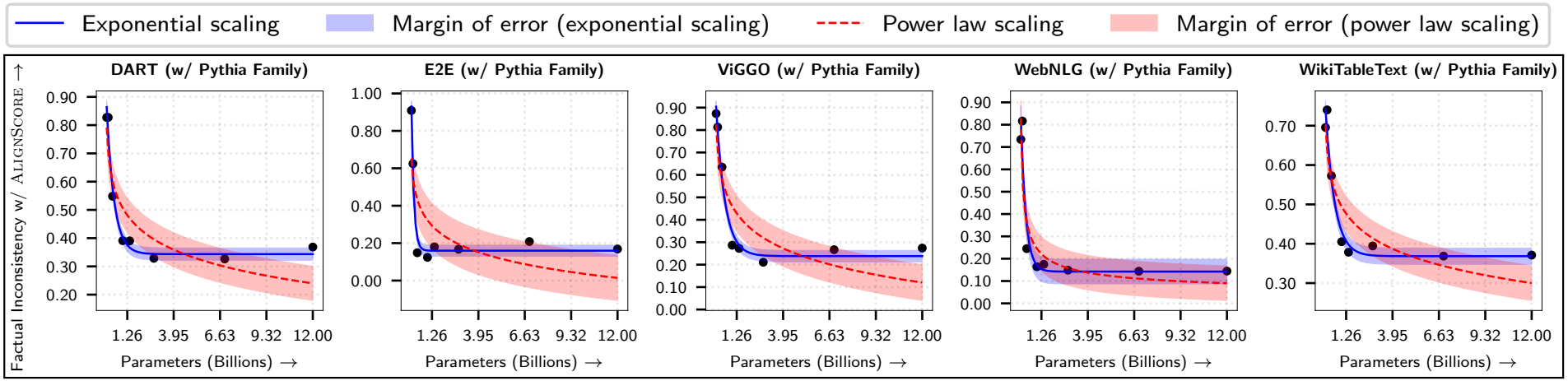


Figure: Fitted power law and exponential scaling models for factual inconsistency, measured by AlignScore.

Table: Results from the *VaCScal* framework.

LLM family	Scaling	DART	E2E	ViGGO	WebNLG	WikiTableText
Pythia	Exponential	2.03E-03/✓	1.71E+01/✓	2.48E-03/✓	2.30E-02/✓	3.31E-03/✓
	Power Law	1.68E+11/✗	1.24E-02/✓	1.92E-02/✗	8.71E+11/✗	1.96E+04/✗

Mathematical Rationale

Assume a D2T dataset exhibiting source-reference divergence, and let factual inconsistency be quantified using the relative perplexity of an LLM trained on this dataset. Then,

$$\text{Factual Inconsistency} \propto e^{-\xi(N)}$$

where $\xi(N)$ is a linear or sub-linear function of the model (LLM) size N.

Conclusion and Future Directions

Conclusion

- Structural representation remains essential.
- Source–reference divergence is a key source of unfaithfulness.
- Selective generation improves faithfulness.
- Factual inconsistency scales exponentially.

Future Directions

- Factual consistency with respect to real-world facts.
- Factual consistency under pre-training and fine-tuning regimes.
- Extending scaling analyses to other generative properties, such as diversity, controllability, and stylistic variation.